

Use Case Name:	Administer System
Scenario:	The WAH4PCE Administrator accesses the monitoring dashboard to review system health, monitor active transaction logs, and manage any data payloads that were quarantined due to missing mandatory fields.
Triggering Event:	The WAH4PCE Admin logs into the ADAPT administrative dashboard.
Brief Description:	This use case allows system administrators to oversee the interoperability pipeline. It provides real-time metrics on successful Local Health Information Exchange (LHIE) transfers and allows manual intervention for quarantined records to prevent data pollution.
Actors:	Primary: WAH4PCE Admin
Related Use Cases:	None
Stakeholders:	WAH4PCE Admin: Needs clear visibility into system traffic and API rate limits. Provider/Requestor: Depends on the admin to resolve blocked or quarantined transfers.
Preconditions:	The WAH4PCE Admin must be authenticated via secure login (e.g., Multi-Factor Authentication).
Postconditions:	System metrics are viewed, and any resolved quarantined records are either purged or pushed back into the processing queue.
Flow of Activities:	Actor: - Admin accesses the dashboard and navigates to the Logs/Quarantine section. - Admin selects a quarantined record for manual review. - Admin resolves the error (e.g., inputs missing PhilHealth ID) and approves retry. System:

	2. System displays real-time metrics (Success Rate, Pending, Quarantined). 4. System safely displays the structural error without exposing full sensitive PHI. 6. System re-queues the payload for processing.
Exception Conditions:	E1: Unauthorized Access: If authentication fails, the system immediately locks the interface and logs a security alert.

Use Case Name:	Request Health Data
Scenario:	An external healthcare facility (like an iHOMIS endpoint) requests the medical history of a referred patient from the WAH network. The system fetches and securely delivers the standardized data.
Triggering Event:	The Requestor initiates an API query to the WAH4PCE gateway for a specific patient's records.
Brief Description:	This use case defines how external endpoints pull standardized health data from the network. It ensures that data leaving the WAH ecosystem is properly structured and securely transmitted to authorized receiving hospitals.
Actors:	Primary: Requestor
Related Use Cases:	None
Stakeholders:	Requestor: Requires accurate, structurally compliant FHIR R4 data to populate their local EMR. Patient: Requires their data to be transmitted securely to their new attending physician.
Preconditions:	Requestor must be a whitelisted endpoint with valid OAuth 2.0 / API credentials.
Postconditions:	The standardized health record is successfully transmitted to the Requestor, and the transaction is logged.
Flow of Activities:	Actor: 1. Requestor sends a secure GET request for patient data. System: 2. System authenticates the endpoint and queries the database. 3. System packages the retrieved data as a PH Core FHIR R4 resource. 4. System transmits the payload to the Requestor via TLS 1.3 encryption.

Exception Conditions:	E1: Patient Not Found: If no record matches the query, the system returns a 404 Not Found error. E2: Unencrypted Connection: If the request is made over HTTP instead of HTTPS, the system immediately terminates the handshake.
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Use Case Name:	Check Request Status
Scenario:	After initiating a data exchange, a Requestor pings the system to see if the transaction is pending, successful, or quarantined/failed.
Triggering Event:	The Requestor submits a specific Transaction ID to the status endpoint.
Brief Description:	This allows external systems to track the progress of their data transfers asynchronously, ensuring they are not left waiting if a payload is quarantined for structural errors.
Actors:	Primary: Requestor
Related Use Cases:	None
Stakeholders:	Requestor: Needs immediate feedback on data transfer status.
Preconditions:	The Requestor must have a valid Transaction ID generated from a previous request/provision of data.
Postconditions:	The current status of the transaction is accurately returned to the actor.
Flow of Activities:	Actor: 1. Requestor sends a status query containing the Transaction ID. System: 2. System searches the sanitized transaction logs for the ID. 3. System returns the status (e.g., "Success", "Pending", "Quarantined").
Exception Conditions:	E1: Invalid Transaction ID: If the ID does not exist, the system returns an error code without revealing any network activity details.

Use Case Name:	Provide Health Data
Scenario:	A health data Provider submits patient records or clinical data into the WAH4PCE Interoperability platform. Depending on the format of the incoming data, it may trigger an AI agent to translate it into standardized formats, which subsequently triggers a mandatory privacy consent validation.
Triggering Event:	The Provider initiates a data transfer, upload sequence, or API call from their local system to the WAH4PCE integration endpoint.
Brief Description:	This use case defines the primary process by which a Provider supplies health data to the ecosystem. It manages core data ingestion while seamlessly handling conditional data translation and strict privacy consent verification.
Actors:	Primary: Provider Secondary: WAH4PCE AI
Related Use Cases:	Extended by: Translate Data to/from PH Core HL7 FHIR R4/HL7 v2
Stakeholders:	Provider: Requires a secure, reliable, and frictionless data transmission pathway. Requestor: Depends on the availability, standardization, and accuracy of the provided data. Patient (Implicit): Requires their health data to be handled securely according to their privacy consent.
Preconditions:	1. Provider must be authenticated and authorized to access the integration endpoint. 2. The WAH4PCE interoperability platform and AI translation services must be online.
Postconditions:	Health data payload is standardized, patient privacy consent is verified, an audit trail is logged, and the data is securely committed to the interoperable repository.
Flow of Activities:	Actor:

	1. Provider initiates the transaction by submitting a health data payload to the gateway. System: 2. System receives the payload, encrypts it in transit, and performs initial structural validation. 3. System analyzes the incoming data format. If the data is non-standard, the system hits an extension point. 4. (Extension) WAH4PCE AI intercepts the workflow to translate the data. 5. (Inclusion) The system triggers the privacy consent validation. 6. If consent is valid, the system commits the standardized data to the repository. 7. System generates and transmits a success status notification back to the Provider.
Exception Conditions:	E1: Corrupted Payload: If the system detects a structurally invalid payload at Step 2, it aborts the transfer and returns a 400 Bad Request error. E2: AI Translation Failure: If the AI encounters anomalous data it cannot map, the transaction is halted, quarantined, and an alert is dispatched. E3: Consent Denied: If the consent check fails, the system instantly purges the payload from active memory and notifies the Provider of a consent-based denial.

Use Case Name:	Translate Data to/from PH Core HL7 FHIR R4/HL7 v2
Scenario:	The WAH4PCE AI automatically intercepts a non-standard data payload and utilizes its Model Context Protocol (MCP) to map and translate the legacy data into the national PH Core FHIR standard.
Triggering Event:	The Provide Health Data use case hits its extension point when it detects an incoming payload that is not in the required FHIR format.
Brief Description:	This is the core microservice function. It acts as an extension to data provision, utilizing AI to dynamically parse and transform unstructured or legacy health data into standardized resources without human intervention.
Actors:	Primary: WAH4PCE AI
Related Use Cases:	Extends: Provide Health Data Includes: Validate Privacy Consent
Stakeholders:	WAH4PCE AI: Requires clear prompt instructions and valid API tokens. Provider: Relies on the AI to prevent them from having to manually re-type data.
Preconditions:	The incoming payload must be successfully intercepted by the microservice, and the AI API (Gemini) must be online.
Postconditions:	A valid JSON payload strictly mapped to the PH Core FHIR data dictionaries is generated and passed back to the main workflow.
Flow of Activities:	Actor (AI): 2. WAH4PCE AI processes the payload text using In-Context Learning. 4. AI maps identified fields to FHIR models (e.g., Vitals to 'Observation'). System: 1. System triggers the extension point and routes data to the MCP server.

	3. System invokes the <<include>> Validate Privacy Consent use case. 5. System receives the translated FHIR resource and closes the extension.
Exception Conditions:	E1: API Rate Limit Hit: If the primary AI model returns a 429 Rate Limit error, the Go middleware executes "Model Juggling" to instantly route the payload to a fallback model.

Use Case Name:	Validate Privacy Consent
Scenario:	Before the AI is allowed to fully process or commit a translated health record, the system automatically checks the payload's metadata to ensure the patient has legally consented to cross-facility data transport.
Triggering Event:	Explicitly invoked by the Translate Data use case via an <<include>> relationship.
Brief Description:	This use case acts as the automated legal gatekeeper. It enforces Data Privacy Act compliance by ensuring that no data is translated or moved without the presence of a valid digital consent flag.
Actors:	Primary: System (Automated)
Related Use Cases:	Included by: Translate Data to/from PH Core HL7 FHIR R4/HL7 v2
Stakeholders:	Patient: Their legal right to data privacy is enforced automatically. WAH4PCE Admin: Protected from legal liabilities regarding unauthorized data sharing.
Preconditions:	The translation use case has begun processing a payload containing metadata.
Postconditions:	The consent flag is validated as "True," allowing the parent translation process to continue.
Flow of Activities:	System (Automated): 1. The parent use case invokes the validation check. 2. System isolates the metadata header of the incoming payload. 3. System scans for the Consent_Flag attribute. 4. System verifies the flag is set to "True" and legally valid for cross-facility transport. 5. System returns a "Proceed" signal to the parent use case.

Exception Conditions:	E1: Missing or False Consent: If the flag is missing, expired, or marked "False," the system instantly aborts the validation, purges the payload from active memory (Ephemeral Processing), and returns an error to the Provider.
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